Mapping and Predicting Foot-and-Mouth Disease Risk in KwaZulu-Natal Using GIS and Machine Learning

Written by Benjamin Klass

1. Introduction

Foot-and-mouth disease (FMD) is a widespread livestock disease prevalent across much of Africa, Asia, and South America. It is often transmitted through wildlife–livestock interaction and results in the weakening and reduced productivity of infected animals. The disease can spread via animal movement as well as through contaminated surfaces. Large-scale FMD epidemics can lead to extensive quarantining and culling of livestock, with severe consequences for already vulnerable rural economies.

South Africa is particularly vulnerable to FMD due to its significant cattle economy, extensive rural communities, and proximity to large wildlife conservation areas. The most affected province is KwaZulu-Natal (KZN), which is characterised by high rural population density and proximity to African buffalo, the primary FMD reservoir in southern Africa. FMD prevalence in the province has increased significantly in recent years and has spread to other provinces.

Relevant scientific literature on foot-and-mouth disease (FMD) occurrence and machine learning–based disease modelling was consulted to inform this investigation.

Mahdi Nazari Ashani et al. (2025) conducted a cross-sectional study examining the socioeconomic and environmental determinants of FMD incidence in Iran using spatial modelling techniques.

Mamadou Ciss et al. (2023) analysed transboundary livestock trade networks in Senegal to assess patterns of animal movement and their implications for FMD transmission across seasons.

Elise Dion, Louis Van Schalkwyk and Eric F. Lambin (2011) developed a spatially explicit multi-agent simulation to represent the spatio-temporal dynamics of FMD transmission at the wildlife–livestock interface near protected areas.

Lina González Gordon et al. (2022) conducted a systematic review of spatial and spatio-temporal studies of FMD outbreaks, highlighting methodological variation and identifying key epidemiological risk factors in endemic settings.

Christian Gortázar et al. (2022) reviewed the role of wildlife as indicators, reservoirs, and transmission hosts in the epidemiology and control of FMD.

Ferran Jori and Eric Etter (2016) developed a stochastic model to estimate the annual risk of FMD transmission from buffalo to cattle at the Kruger National Park wildlife–livestock interface.

Virgilius Martin Kelake Kedang, Indri Permatasari and Thanicha Chanchaidechachai (2025) performed spatial and spatio-temporal analyses to identify outbreak patterns and associated risk factors in Java Island between 2022 and 2023.

Bachir Souley Kouato et al. (2018) reviewed epidemiological modelling approaches for FMD, identifying commonly used methods, key risk factors, and limitations across studies.

Xiaoxiao Li et al. (2021) developed a space-time framework for generating susceptibility maps for infectious diseases, providing methodological guidance relevant to spatial risk modelling.

Oscar Madzingira, Michael Sinvula Lukubwe and Evans Simasiku (2025) examined the implications of the wildlife–livestock interface for ecological sustainability, livestock production systems, and rural livelihoods.

Iain McLachlan et al. (2019) applied compartmental modelling to investigate FMD persistence and outbreak dynamics within smallholder cattle herds in endemic regions.

Oriane Ploquin et al. (2025) analysed spatial patterns of FMD exposure and associated risk factors in livestock populations located near wildlife areas in Zimbabwe using longitudinal serological data.

Budina Eka Prasetia, Roza Azizah Primatika and Widagdo Sri Nugroho (2025) conducted a GIS-based spatial risk assessment using the Analytical Hierarchy Process (AHP) to identify high-risk areas for FMD in Sukabumi Regency.

Qi An et al. (2025) developed an ensemble ecological niche model incorporating climatic, geographic, and socio-economic variables to predict FMD suitability and generate species-specific risk maps.

Fanos Tadesse Woldemariyam et al. (2023) reviewed the epidemiological dynamics of FMD in the Horn of Africa, focusing on viral diversity, livestock production systems, animal movement, and the role of wildlife in disease persistence.

The aim of this study is to identify the key factors influencing the distribution and spread of FMD in KwaZulu-Natal, South Africa.

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2. Method

The objective of this investigation was to assess the influence of road density, proximity to the Disease Management Area (DMA), and proximity to protected areas on FMD occurrence at the local municipality level in KwaZulu-Natal.

The primary spatial dataset was derived from a South African government FMD technical report (Department of Agriculture, Land Reform and Rural Development, 2025) which included a national map of active, historical, and closed FMD cases. The KwaZulu-Natal portion of this map was extracted and georeferenced using identifiable anchor points near major settlements, resulting in a high-accuracy spatial reference.

A point layer of FMD infection locations was then manually digitised from the georeferenced map. Each point was classified according to its status (2025, pre-2025, or closed), allowing for differentiated visualisation.

The number of infection points per municipality was calculated using a spatial aggregation (point-in-polygon count), enabling the determination of both total cases and case density.

A South African administrative boundaries dataset was obtained from OpenStreetMap via the Humanitarian Data Exchange (Humanitarian OpenStreetMap Team, 2026). From this dataset, the provincial boundary of KwaZulu-Natal and its constituent local municipalities were extracted and clipped for use in subsequent spatial analysis.

A Disease Management Area (DMA) layer was created based on municipalities listed in the technical report. Only fully included municipalities were incorporated, as partial municipality delineation was not feasible within the time constraints.

A national road network shapefile was obtained from the Humanitarian Data Exchange (Humanitarian OpenStreetMap Team, 2026). Road density was calculated by summing total road length within each municipality and dividing by municipal area.

Municipalities were classified according to DMA membership (1 = inside, 0 = outside). Municipal centroids were generated to represent geographic midpoints, and distances to the DMA boundary were calculated by converting the DMA polygon into a line feature and measuring centroid-to-boundary distance.

Similarly, distances to key FMD reservoir protected areas—Hluhluwe–iMfolozi and iSimangaliso—were calculated.

All predictor variables (road density, DMA distance, protected area distance) were normalised to a 0–1 scale and combined into a composite risk index.

For machine learning analysis, a binary target variable representing outbreak presence was created (1 = municipality contains cases, 0 = no cases). The dataset was exported to CSV and analysed in Python using a Random Forest classifier. A 70:30 train-test split was applied, and model accuracy and feature importance were calculated.

3. Results

Model performance:

- Accuracy: **0.71**
- Precision (no outbreak): 0.60
- Precision (outbreak): 0.78

This indicates moderate predictive performance, with stronger identification of outbreak-positive municipalities.

	precision	recall	f1-score	support
0	0.60	0.60	0.60	5
1	0.78	0.78	0.78	9
accuracy			0.71	14
macro avg	0.69	0.69	0.69	14
weighted avg	0.71	0.71	0.71	14

Figure 1: Model performance

Feature importance:

- DMA proximity: **0.40**
- Protected area proximity: **0.35**
- Road density: **0.23**

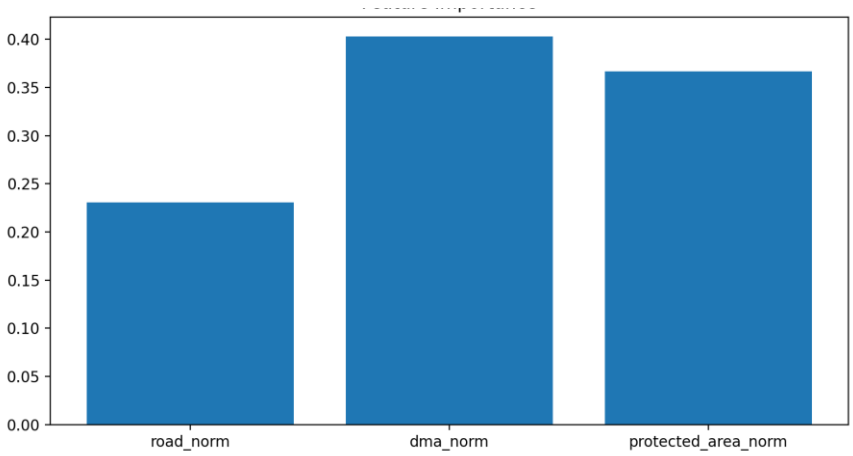

Figure 2: Feature importance

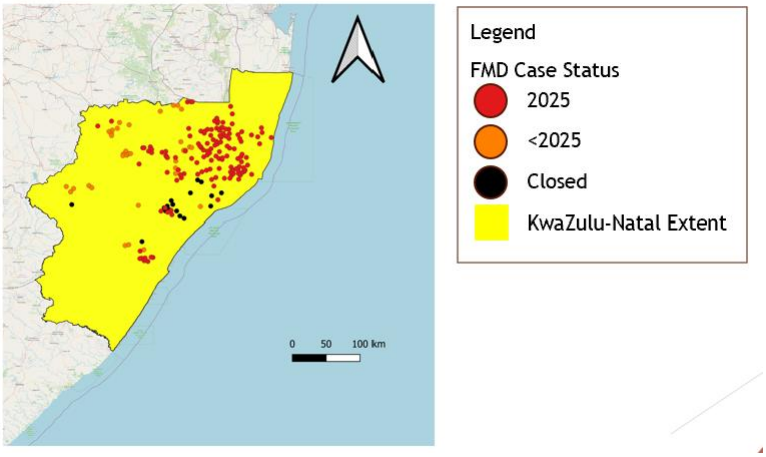

Figure 3: KZN FMD Cases Map

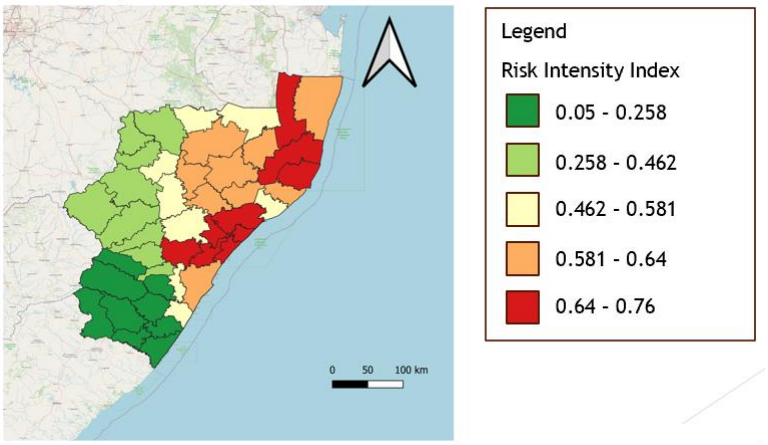

Figure 4: KZN Risk Intensity Map

4. Discussion

Municipalities with the highest FMD occurrence are generally those located within or near the DMA, or those containing or bordering the Hluhluwe–iMfolozi and iSimangaliso protected areas. As distance from these features increases, FMD occurrence tends to decrease.

KwaZulu-Natal's road network is densest along the central and southern coast, particularly along the Durban–Pietermaritzburg corridor. This network likely facilitates the spread of FMD beyond the northern source areas.

The composite risk index is highest in northeastern municipalities near the DMA and protected areas, but the highest individual score occurs in KwaDukuza Local Municipality. This highlights the importance of transport connectivity, as this municipality lies along a major movement corridor between the northern outbreak zone and the central economic region.

The municipalities with the highest risk index—KwaDukuza, uMlalazi, Ndwedwe, Mandeni, and uMshwathi—are concentrated either along the northern wildlife interface or along the central coastal transport corridor.

A methodological limitation is that distances were calculated from municipal centroids rather than boundaries, which may not fully capture intra-municipal spatial variation. Additionally, care had to be taken to avoid feature duplication caused by improper joins, ensuring that the final dataset correctly represented the 44 municipalities of KwaZulu-Natal.

Feature importance analysis indicates that DMA proximity and protected area proximity are the dominant predictors of outbreak occurrence, while road density plays a secondary role. This supports a model in which ecological and regulatory factors drive initial disease emergence, while transport networks facilitate subsequent spread.

Given the relatively small sample size ($n \approx 44$), the model should be interpreted as identifying spatial associations rather than providing fully generalisable predictions.

5. Conclusion

This investigation aimed to assess the influence of road density, DMA proximity, and protected area proximity on FMD occurrence in KwaZulu-Natal.

The results indicate that DMA proximity is the strongest predictor of outbreak occurrence, followed by proximity to protected areas and road density. Two main clusters of FMD risk

were identified: a primary cluster in the north associated with wildlife reservoirs and the DMA, and a secondary cluster along the central coast associated with high road density and transport connectivity.

These findings support a two-stage disease system in which FMD originates in northern municipalities through wildlife–livestock interaction and subsequently spreads southwards along major transport routes.

This highlights the need for a dual intervention strategy: strengthening wildlife–livestock separation and control measures within the DMA, while also limiting disease spread through improved regulation of livestock movement along key transport corridors.

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6. References

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7. Geospatial Layers and Code

7.1 QGIS Layers

The following geospatial layers were used in the FMD spatial analysis:

a. FMD cases
Vector dataset representing recorded FMD outbreak locations across KwaZulu-Natal (KZN), categorized by outbreak status.

b. Road network
Line dataset representing the major road infrastructure within KZN.

c. Protected area boundaries
Polygon dataset delineating key FMD-associated protected areas, including Hluhluwe–iMfolozi Park and iSimangaliso Wetland Park.

d. Municipality centroids
Point dataset representing the geographic centroids of KZN local municipalities.

e. KZN boundary map
Polygon dataset of KwaZulu-Natal, clipped from the national administrative boundaries dataset derived from OpenStreetMap.

f. Local municipalities
Polygon dataset representing the spatial extents of local municipalities within KZN.

g. Road density
Derived layer representing road length normalized by municipal area, indicating infrastructure intensity per municipality.

h. DMA distance
Derived layer representing the distance between each municipality and the nationally designated FMD Disease Management Area (DMA).

i. Protected area distance
Derived layer representing the distance between municipalities and major FMD-associated protected areas (Hluhluwe–iMfolozi and iSimangaliso).

j. Risk intensity index
A composite layer was developed to estimate the relative risk of FMD occurrence across municipalities by integrating three normalized variables: road density, distance to the DMA, and distance to protected areas. The index was calculated as:

R_i = (1/3)(RD_i + D_{DMA,i}^* + D_{PA,i}^*)

where R_i represents the risk intensity for municipality *i*, RD_i is the normalized road density, D_DMA,i* is the normalized inverse distance to the DMA, and D_PA,i* is the normalized inverse distance to protected areas.

Distance variables were transformed using min–max normalization and subsequently inverted so that municipalities located closer to the DMA and protected areas received higher risk scores. All variables were equally weighted, such that $w_1 = w_2 = w_3 = 1/3$, ensuring a balanced contribution of each factor. Higher index values therefore indicate greater relative FMD risk.

7.2 Python Code

The following Python workflow was used to implement a machine learning classification model for FMD outbreak prediction:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("KZN_FMD_dataset.csv")
print(df.head())

# Define target variable
y = df["outbreak_present"]

# Define feature variables
X = df[["road_norm", "dma_norm", "protected_area_norm"]]

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
```

```
)

# Initialize Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Generate predictions
y_pred = model.predict(X_test)

# Evaluate model performance
print(classification_report(y_test, y_pred))

# Assess feature importance
importances = model.feature_importances_
features = X.columns

plt.bar(features, importances)
plt.title("Feature Importance")
plt.xlabel("Features")
plt.ylabel("Importance Score")
plt.show()
```